MIL-T-48149A(PA)
14 January 1976
SUPERSEDING
MIL-T-48149(MU)
3 April 1973

MILITARY SPECIFICATION

TITANIUM-DIOXIDE; DRY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers titanium dioxide (TiO_2) intended for use in additive liners (see 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

FEDERAL

FED-STD-141	- Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1234	- Pyrotechnics: Sampling, Inspection and Testing.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC: 6810

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D153	- Specific Gravity of Pigments, Standard Methods of Test for.
ASTM D185	- Coarse Particles in Pigments, Pastes and Paints, Standard Methods of Test for.
ASTM D280	- Hygroscopic Moisture (and Other Matter Volatile Under the Test Conditions) in Pigments, Standard Methods of Test for.
ASTM D281	- Oil Absorption of Pigments by Spatula Rub-Out, Standard Methods of Test for.
ASTM D1394	- Chemical Analysis of White Titanium Pigments, Standard Methods for.
ASTM E70	- pH of Aqueous Solution with the Glass Electrode, Test for.
ASTM E300	- Sampling Industrial Chemicals, Standard Recommended Practice for.

3. REQUIREMENTS

3.1 Titanium-dioxide.-The titanium dioxide content shall be 98.0 percent minimum, when determined as specified in 4.5.1.

3.2 Coarse Particles.-Not over 0.25 percent shall be retained on U.S. Standard number 325 sieve, when tested as specified in 4.5.2.

3.3 Specific Gravity.-The specific gravity shall be between 3.8 and 4.2 at 15.6 degrees Centigrade (°C), when tested as specified in 4.5.3.

3.4 Oil Absorption.-The oil absorption shall be between 20.0 and 40.0 pounds per 100 pounds of titanium dioxide when determined as specified in 4.5.4. The results of the test shall be furnished to the government in accordance with data item description DI-R-1724 as specified in the contract (see 6.6).

3.5 Water Solubility.-The water solubility shall not exceed 0.2 percent, when determined as specified in 4.5.5.

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3.6 Hydrogen ion Concentration.-The pH shall be between 7.0 and 8.0, when determined as specified in 4.5.6.

3.7 Moisture.-The moisture content shall not exceed 0.7 percent, when determined as specified in 4.5.7.

3.8 Compatibility.-The gas evolved shall not exceed three (3) millilliters when determined as specified in 4.5.8. The results of the test shall be furnished to the government in accordance with data item description DI-R-1724 as specified in the contract (see 6.6).

3.9 First Article Inspection.-This specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspections shall be conducted on this item:

a. First Article Inspection (see 4.3).
b. Quality Conformance Inspection (see 4.4).

4.3 First Article Inspection.

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of two (2) pounds of titanium dioxide obtained by sampling as described in 4.4.3. The sample shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be Performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.5 of this specification.

4.3.3 Rejection.-See MIL-A-48078.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. For the material covered by this specification, a lot shall consist of one or more batches of titanium dioxide produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of titanium dioxide that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.4.2 Examination.-Unless otherwise specified in the Classification of Defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II (see MIL-A-48078).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Container, Sealed		NA
			NEXT HIGHER ASSEMBLY: NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major B:					
131.	Container damaged to extent that contents are exposed or liable to become exposed.		0.40%	5.1	Visual
132.	Marking incorrect, misleading or unidentifiable.		0.40%	5.1	Visual
Minor:	None defined.				

NOTES:

SARPA-QA FORM 2567 FEB 74

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4.4.3 Testing.

4.4.3.1 Sampling.-The tests depicted in 4.5.1 through 4.5.8 shall be performed on a representative sample from each batch. Approximately one pound of titanium dioxide shall be selected from each batch in accordance with ASTM E300 for solids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, AQL 6.5 percent, using a batch as a unit of product. If any sample fails to meet any requirement, the batch represented by the sample shall be rejected. In addition, after any failure of a batch the contractor will return to 100 percent inspection until "i" successive batches are accepted as required by MIL-STD-1235. The classification for the tests shall be as given in Table I.

Table I

Inspection/Requirement	Defect Classification	Test Paragraph
TiO_2 Content (see 3.1)	Major B	4.5.1
Coarse Particles (see 3.2)	Major B	4.5.2
Specific Gravity (see 3.3)	Major B	4.5.3
Oil Absorption (see 3.4)	Major B	4.5.4
Water Solubility (see 3.5)	Major B	4.5.5
Hydrogen ion Concentration (see 3.6)	Major B	4.5.6
Moisture (see 3.7)	Major B	4.5.7
Compatibility (see 3.8)	Major B	4.5.8

4.4.4 Inspection Equipment.-The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment shall be employed where applicable for all tests and examinations specified in 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.4).

4.5 Test Methods and Procedures (see 6.5).

4.5.1 Titanium Dioxide Content.-The titanium dioxide content shall be determined in accordance with ASTM-D1394.

4.5.2 Coarse Particles.-The amount of coarse particles shall be determined in accordance with ASTM-D185.

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4.5.3 Specific Gravity.-The specific gravity shall be determined in accordance with ASTM-D153.

4.5.4 Oil Absorption.-The oil absorption shall be determined in accordance with ASTM-D281.

4.5.5 Water Solubility.-The water solubility content shall be determined in accordance with FED-STD-141, Method 5241. Whatman #3 or #44 are acceptable alternates to the filter paper stated in FED-STD-141, Method 5241.

4.5.6 Hydrogen ion Concentration.-Accurately weigh a 5 gram specimen and place it in a glass stoppered Erlenmeyer flask. Add 50 milliliters of freshly boiled and distilled water to the flask. Stopper the flask and shake well. Cool the mixture to 25°C, while occasionally shaking the stoppered flask. Determine the pH of the clear aqueous layer in accordance with ASTM-E70. Determine the pH within a period of not more than 10 minutes after exposure of the extract to air.

4.5.7 Moisture Content.-The moisture content shall be determined in accordance with ASTM-D280, Method A.

4.5.8 Compatibility.-The compatibility shall be in accordance with MIL-STD-1234, Method 504.1, except that M30 propellant shall be used in place of a pyrotechnic. The propellant shall be rasped or cut and then passed through number 12 and 20 sieve and the material retained on the number 20 sieve shall be mixed to form a homogeneous sample to be used in the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-Unless otherwise specified, commercial packaging will be acceptable.

5.2 Packing.-The titanium dioxide shall be prepared for shipment in such a manner as to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.3 Marking.-Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 The titanium dioxide covered by this specification is intended for use as an ingredient in the composition (TiO_2/Wax) used to manufacture additive liners for artillery cartridges.

6.2 Sources of titanium dioxide which have been found satisfactory for intended use are:

1) Ti-Pure LW Grade, Dupont, Wilmington, Delaware.
2) Unitane 0-220, American Cyanamid, Pigment Div, Wayne, New Jersey.
3) Titanox 2005, National Lead, South Amboy, New Jersey.
4) Tronox CR833, Kerr McGee, Oklahoma City, Oklahoma.

The Dupont Ti-Pure LW and the American Cyanamid Unitane 0-220 are preferred on the basis of better incorporation with wax and less scrap in liner manufacture.

Use of any other materials other than the four listed above shall require prior approval by Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-A-P.

6.3 Ordering Data.-See MIL-A-48078.

6.4 Submission of Inspection Equipment Designs for Approval.-See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

6.5 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Data Requirements.-Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
3.4	Test results for oil absorption	DI-R-1724 (Preparation instructions 1 and 2 only, should apply).

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Paragraph	Data Requirement	Applicable DID
3.8	Test results for compatability	DI-R-1724 (Preparation instructions 1 and 2 only, should apply).

(Copies of data item descriptions required by the contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer. Distribution of data required by DI-R-1724 shall include Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801).

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 6810-AA73

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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